

With such an obsession around solving puzzles, it's obvious that unsolved puzzles are by far the most popular, or the ones that capture our imagination. Over the years, there have been many such unsolved puzzles which have caught the imagination of the masses. The notion of invincibility of a cipher has led millions to try their hands at solving them. Despite tremendous advances in the science of ciphers, and using high-end and complex decryption techniques, there are several ciphers still waiting to be translated. In the modern world, where information is power, the field of data encryption has acquired a mighty status. Here we look at some of the famous unbreakable ciphers that have taunted us for years.

One-time Pad

One-time pad, sometimes known as the perfect cipher, is an encryption algorithm which combines plaintext with a random key to generate a cipher. Provided that the usage is correct, this type of encryption is the only proven unbreakable encryption. Invented in 1917, the one-time pad was derived from the Vernam cipher, which was named in honor of one of its inventors, Gilbert Vernam. The Vernam cipher used a tape for encryption which was looped after each cycle and reused thereby increasing its vulnerability. Owing to its usage by Special Ops teams in World War, intelligence agencies and spies during the Cold War and also in protecting diplomatic communication, the one-time pad gained a reputation for being a simple, yet effective encryption system.



One-time Pad

Each character from the plaintext is encrypted by a modular addition with a character from a secret random key of the same length as the plain text, resulting in a cipher text. The cipher text is impossible to decrypt provided that the key is random, with same or larger length than the plaintext, never reused again and kept secret (obviously).